Lecture Activity: Grasshopper Lengths

Learning Goals:

- Become more familiar with summary statistics in JASP
- Practice calculating and interpreting summary statistics

Getting Started:

You have randomly sampled some grasshoppers from two separate fields nearby campus and measured their length in centimeters.

Download the CSV file of their lengths from here (https://umbc.box.com/s/0p1jzz7114nwgcybt1lyozew06n7aci4), and open it in JASP.

If you cannot access JASP due to technical difficulties, please ask your instructor for the alternative method of completing the assignment.

Tasks:

To do in JASP:

- For each field, find
  - The mean and median grasshopper length from your samples
  - The variance and standard deviation of grasshopper lengths from your samples
  - The standard error of the mean for your samples
  - The 95% confidence interval from your samples

Questions to answer:

- Your sample set from which field has observations that generally lie further from the mean? How can you tell?
- Your sample set from which field has a more precise estimate of the population mean? How can you tell?
- Your sample from field A has a couple of outliers that could throw things off - however, ignoring that, what's one good reason that Field B has a smaller standard error?
- Pick the confidence interval from your sample from either Field A or Field B and give a one-sentence interpretation of it.
- Your lab assistant Guy Fieri says that he looked at grasshoppers from Field B and measured their lengths. He tells you that the mean grasshopper length in Field B is 8.5 centimeters long!

Do you think this is plausible, or do you want to take another look at the bugs he measured? Why?

Deliverables:

- A PDF or screenshot of the JASP results, uploaded within the assignment portal. (Worth 5 points)
- Your answers to the questions above, submitted either as text in the assignment portal or uploaded as a file. (Worth 5 points, 1 for each question)